

Harsh Vassaram

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SUMMARY

Computer Science Graduate, ASU (GPA: 3.74) with hands-on research experience in RAG systems. Redesigned and shipped a production-ready frontend in React, Vite, and TypeScript — translating Figma prototypes into a deployed capstone platform, achieving a 20% improvement in user workflow. Experienced in benchmarking and integrating LLM APIs across domain-specific RAG pipelines. Multilingual communicator with a track record of delivering data-driven results in cross-functional environments.

EDUCATION

Bachelor of Science in Computer Science
Arizona State University - Tempe, AZ

May 2026
GPA: 3.74

Associate of Science in Computer Science
Mesa Community College - Mesa, AZ

Dec 2023

TECHNICAL SKILLS

- **Languages:** Python, C/C++, C#, Java, Typescript, JavaScript
- **Frameworks & Backend:** React, Vite, .NET, REST APIs
- **System & Cloud:** Linux, Distributed Systems, Microsoft Azure, GitHub Actions, CI/CD
- **AI / ML:** Retrieval-Augmented Generation (RAG) Systems, Vector Databases (Milvus), LLM APIs, Neural Networks
- **Tools & Design:** Git, VS Code, Figma, PostgreSQL

PROJECTS

Fulton Undergraduate Research Initiative - Chunking Strategies for RAG

Spring 2026

- Engineered and benchmarked different chunking strategies (fixed, semantic, hybrid) in RAG pipelines.
- Analyzed domain-specific data sources to tailor chunking strategies, reducing model "hallucinations".
- Evaluated computational cost vs. response accuracy trade-offs to achieve high-fidelity generation.

Viking Roots - AI-Powered Heritage Storytelling Platform (Capstone)

Fall 2025 - Spring 2026

- Redesigned the frontend UI using React, Vite, and Typescript via Figma prototypes, improving user workflow navigation by **20%** based on usability testing.
- Refined the system prompt of the AI questionnaire feature, which used the Gemini API.

Predictive Machine Learning Modeling for Bank Customer Retention

Fall 2025

- Developed an Artificial Neural Network (ANN) with **90% accuracy** in predicting customer subscriptions.
- Crafted a preprocessing pipeline for **45000+ records**, optimizing **17 features** using One-Hot encoding.
- Conducted a bias report on the call duration metrics to improve the model & reduce the risk of losing high-net-worth individuals.

WORK EXPERIENCE

Engineering Dean's Office Operations Assistant

Mar 2024 - May 2026

Ira A. Fulton Schools of Engineering, Arizona State University, Tempe, AZ

- Aided in data analysis & report creation in Excel using Power Query, combining metrics from proprietary data sources to increase faculty reporting efficiency by **30%**.

Front Desk Support

May 2022 - Dec 2023

International Education Office, Mesa Community College, Mesa, AZ

- Collaborated with Keystone Education on a data-driven project to increase student enrollment by **20%**.

LEADERSHIP EXPERIENCE

Fulton Academic Integrity Matters - Student Representative

Mar 2025 - May 2026

Ira A. Fulton Schools of Engineering, Arizona State University

- Represented the student perspective in academic integrity cases, recommending outcomes.

LANGUAGES: English (Fluent), Portuguese (Fluent), Gujarati (Native), Hindi (Fluent)